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Advice for applying machine learning

Deciding what to try next

Debugging a learning algorithm

You've implemented regularized linear regression on housing prices

$$J(\vec{w}, b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)^2 + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2$$

But it makes unacceptably large errors in predictions. What do you try next?

- Get more training examples
- Try smaller sets of features
- Try getting additional features
- Try adding polynomial features $(x_1^2, x_2^2, x_1x_2, etc)$
- Try decreasing λ
- Try increasing λ

Machine learning diagnostic

Diagnostic: A test that you run to gain insight into what is/isn't working with a learning algorithm, to gain guidance into improving its performance.

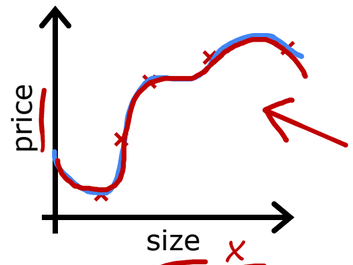
Diagnostics can take time to implement but doing so can be a very good use of your time.



Evaluating and choosing models

Evaluating a model

Evaluating your model



→ Model fits the training data well but will fail to generalize to new examples not in the training set.

$W_n x^4$

$x = x_1$

$$f_{\vec{w},b}(\vec{x}) = w_1 x + w_2 x^2 + \cdots + w_n x^n + b$$

just x

- → x_1 = size in feet^2
- → x_2 = no. of bedrooms
- → x_3 = no. of floors
- → x_4 = age of home in years

$f(\vec{x})$

Evaluating your model

Dataset:

size	price
2104	400
1600	330
2400	369
1416	232
3000	540
1985	300
1534	315
1427	199
1380	212
1494	243

70% training set

30% test set

$(x^{(1)}, y^{(1)})$
$(x^{(2)}, y^{(2)})$
$\vdots$
$(x^{(m_{train})}, y^{m_{train}})$

m_{train} = no. training examples = 7

$(x_{test}^{(1)}, y_{test}^{(1)})$
$\vdots$
$(x_{test}^{(m_{test})}, y_{test}^{(m_{test})})$

m_{test} = no. test examples = 3

Train/test procedure for linear regression (with squared error cost)

Fit parameters by minimizing cost function $J(\vec{w}, b)$

$$J(\vec{w}, b) = \min_{\vec{w}, b} \left[\frac{1}{2m_{train}} \sum_{i=1}^{m_{train}} \left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)} \right)^2 + \frac{\lambda}{2m_{train}} \sum_{j=1}^{n} w_j^2 \right]$$

Compute test error:

$$J_{test}(\vec{w}, b) = \frac{1}{2m_{test}} \left[\sum_{i=1}^{m_{test}} \left(f_{\vec{w},b}(\vec{x}^{(i)}_{test}) - y^{(i)}_{test} \right)^2 \right]$$

Compute training error:

$$J_{train}(\vec{w}, b) = \frac{1}{2m_{train}} \left[\sum_{i=1}^{m_{train}} \left(f_{\vec{w},b}(\vec{x}^{(i)}_{train}) - y^{(i)}_{train} \right)^2 \right]$$

Train/test procedure for linear regression (with squared error cost)

Fit parameters

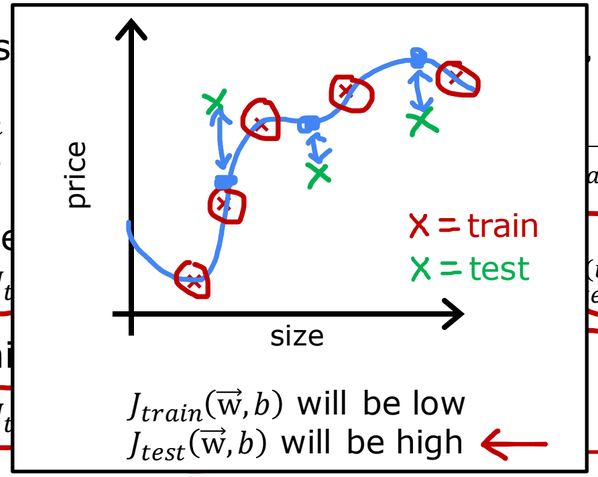
$J(\vec{w}, b) = \min_{\vec{w}, b}$

Compute test e

J_t

Compute traini

J_t



$b)$

$\frac{1}{ain}\left[\sum_{j=1}^{n} w_j^2\right]$

$\left(i\right)_{est}\Big)^2\Big]$ + $\sum w^2$

$-\ y_{train}^{(i)}\Big)^2\Big]$

Train/test procedure for classification problem

0/1

Fit parameters by minimizing $J(\vec{w}, b)$ to find $\vec{w}, b$

E.g.

$$J(\vec{w}, b) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right) + \left(1-y^{(i)}\right)\log\left(1-f_{\vec{w},b}\left(\vec{x}^{(i)}\right)\right)\right] + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2$$

Compute test error:

$$J_{test}(\vec{w}, b) = -\frac{1}{m_{test}}\sum_{i=1}^{m_{test}}\left[y_{test}^{(i)}\log\left(f_{\vec{w},b}\left(\vec{x}_{test}^{(i)}\right)\right) + \left(1-y_{test}^{(i)}\right)\log\left(1-f_{\vec{w},b}\left(\vec{x}_{test}^{(i)}\right)\right)\right]$$

Compute train error:

$$J_{train}(\vec{w}, b) = -\frac{1}{m_{train}}\sum_{i=1}^{m_{train}}\left[y_{train}^{(i)}\log\left(f_{\vec{w},b}\left(\vec{x}_{train}^{(i)}\right)\right) + \left(1-y_{train}^{(i)}\right)\log\left(1-f_{\vec{w},b}\left(\vec{x}_{train}^{(i)}\right)\right)\right]$$

Train/test procedure for classification problem

0/1

Fit parameters by minimizing $J(\vec{w}, b)$ to find $\vec{w}, b$

E.g.,

$$J(\vec{w}, b) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(f_{\vec{w},b}(\right.\right.$$

Compute test error:

$$J_{test}(\vec{w}, b) = -\frac{1}{m_{test}}\sum_{i=1}^{m_{test}}\left[y\right.$$

Compute train error:

$$J_{train}(\vec{w}, b) = -\frac{1}{m_{train}}\sum_{i=1}^{m_{train}}$$

fraction of the test set and the fraction of the train set that the algorithm has misclassified.

$$\hat{y} = \begin{cases} 1 \text{ if } f_{\vec{w},b}(\vec{x}^{(i)}) \geq 0.5 \\ 0 \text{ if } f_{\vec{w},b}(\vec{x}^{(i)}) < 0.5 \end{cases}$$

count $\hat{y} \neq y$

$J_{test}(\vec{w}, b)$ is the fraction of the test set that has been misclassified.

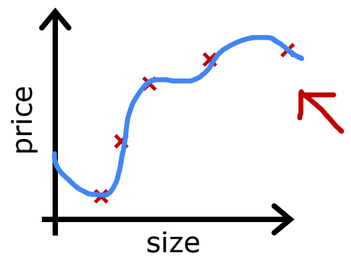
$J_{train}(\vec{w}, b)$ is the fraction of the train set that has been misclassified.



Evaluating and choosing models

Model selection and training/cross validation/test sets

Model selection (choosing a model)



$$f_{\vec{w},b}(\vec{x}) = w_1x_1 + w_2x^2 + w_3x^3 + w_4x^4 + b$$

Once parameters $\vec{w}, b$ are fit to the training set, the training error $J_{train}(\vec{w}, b)$ is likely lower than the actual generalization error.

$J_{test}(\vec{w}, b)$ is better estimate of how well the model will generalize to new data than $J_{train}(\vec{w}, b)$.

Model selection (choosing a model)

1. $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + b$ → $w^{<1>}, b^{<1>}$ → $J_{test}(w^{<1>}, b^{<1>})$
2. $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x^2 + b$ → $w^{<2>}, b^{<2>}$ → $J_{test}(w^{<2>}, b^{<2>})$
3. $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x^2 + w_3 x^3 + b$ → $w^{<3>}, b^{<3>}$ → $J_{test}(w^{<3>}, b^{<3>})$

⋮

10. $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x^2 + \cdots + w_{10} x^{10} + b$ → $J_{test}(w^{<10>}, b^{<10>})$

Choose $w_1 x_1 + \cdots + w_5 x^5 + b$ $d=5$ $J_{test}(w^{<5>}, b^{<5>})$

How well does the model perform? Report test set error $J_{test}(w^{<5>}, b^{<5>})$?

The problem is $J_{test}(w^{<5>}, b^{<5>})$ is likely to be an optimistic estimate of generalization error. Ie: An extra parameter d (degree of polynomial) was chosen using the test set.

w, b

Training/cross validation/test set

size	price
2104	400
1600	330
2400	369
1416	232
3000	540
1985	300
1534	315
1427	199
1380	212
1494	243

validation set
development set
dev set

training set 60%

cross validation 20%

test set 20%

$(x^{(1)}, y^{(1)})$
$\vdots$
$(x^{(m_{train})}, y^{(m_{train})})$

$m_{train} = 6$

$(x_{cv}^{(1)}, y_{cv}^{(1)})$
$\vdots$
$(x_{cv}^{(m_{cv})}, y_{cv}^{(m_{cv})})$

cross validation

$m_{cv} = 2$

$(x_{test}^{(1)}, y_{test}^{(1)})$
$\vdots$
$(x_{test}^{(m_{test})}, y_{test}^{(m_{test})})$

$m_{test} = 2$

Training/cross validation/test set

Training error: $J_{train}(\vec{w}, b) = \frac{1}{2m_{train}}\left[\sum_{i=1}^{m_{train}} \left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)^2\right]$

Cross validation error: $J_{cv}(\vec{w}, b) = \frac{1}{2m_{cv}}\left[\sum_{i=1}^{m_{cv}} \left(f_{\vec{w},b}\left(\vec{x}_{cv}^{(i)}\right) - y_{cv}^{(i)}\right)^2\right]$ (validation error, dev error)

Test error: $J_{test}(\vec{w}, b) = \frac{1}{2m_{test}}\left[\sum_{i=1}^{m_{test}} \left(f_{\vec{w},b}\left(\vec{x}_{test}^{(i)}\right) - y_{test}^{(i)}\right)^2\right]$

Model selection

d为 dimension

d=1 1. $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + b$

d=2 2. $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x^2 + b$

d=3 3. $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x^2 + w_3 x^3 + b$

⋮ ⋮

d=10 10. $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x^2 + \cdots + w_{10} x^{10} + b$

$w^{<1>}, b^{<1>}$

$J_{cv}(w^{<1>}, b^{<1>})$

$J_{cv}(w^{<2>}, b^{<2>})$

⋮

$J_{cv}(w^{<10>}, b^{<10>})$

找 lowest error

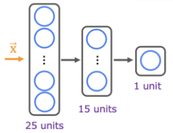
→ Pick $w_1 x_1 + \cdots + w_4 x^4 + b$ $(J_{cv}, w^{<4>}, b^{<4>})$

我们是从 cross validation 报告 → J test的结果

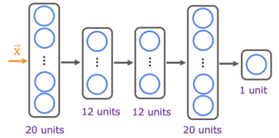
Estimate generalization error using test the set: $J_{test}(w^{<4>}, b^{<4>})$

Model selection – choosing a neural network architecture

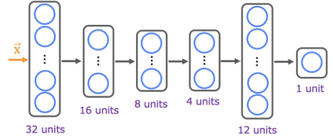
不同模型的选择

1. 

$w^{(1)}, b^{(1)}$ $J_{cv}(\mathbf{W}^{(1)}, \mathbf{B}^{(1)})$

2. 

$w^{(2)}, b^{(2)}$ $J_{cv}(\mathbf{W}^{(2)}, \mathbf{B}^{(2)})$

3. 

$w^{(3)}, b^{(3)}$ $J_{cv}(\mathbf{W}^{(3)}, \mathbf{B}^{(3)})$

用交叉验证集选择模型
用测试集报告误差
以免 train set 容易取得过小

Train, CV

J_{cv} (Cross Validation)

根据 J_{cv} 来报告最后的泛化误差
选择对应的 test set

Pick $\mathbf{W}^{(2)}, \mathbf{B}^{(2)}$

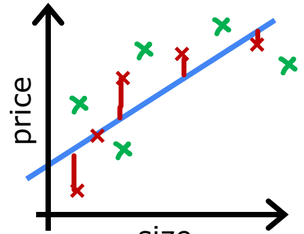
Estimate generalization error using the test set: $J_{test}(\mathbf{W}^{(2)}, \mathbf{B}^{(2)})$



Bias and variance

Diagnosing bias and variance

Bias/variance



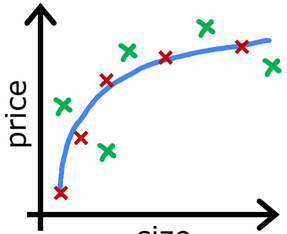
$f_{\vec{w},b}(x) = w_1 x + b$

High bias (underfit)

$d = 1$

J_{train} is high

J_{cv} is high



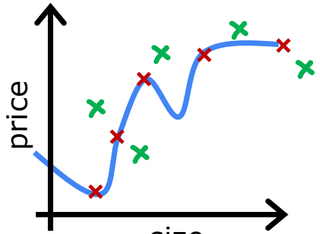
$f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + b$

"Just right"

$d = 2$

J_{train} is low

J_{cv} is low



$f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + b$

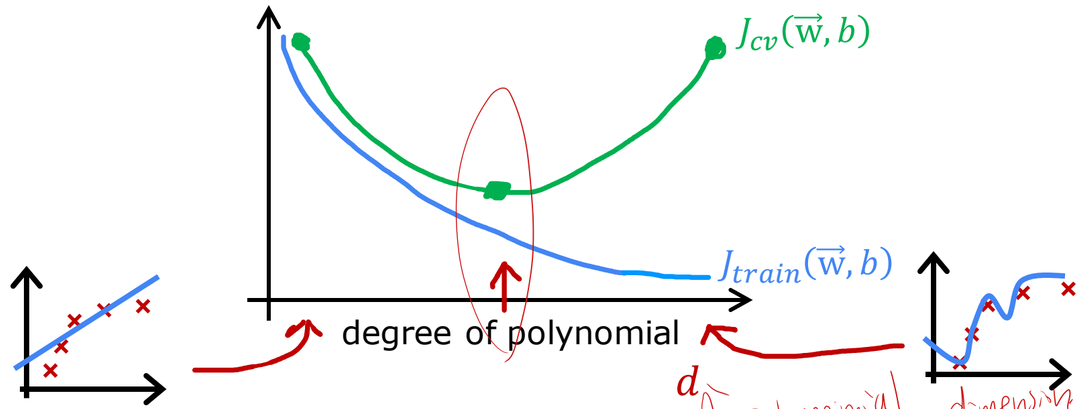
High variance (overfit)

$d = 4$

J_{train} is low

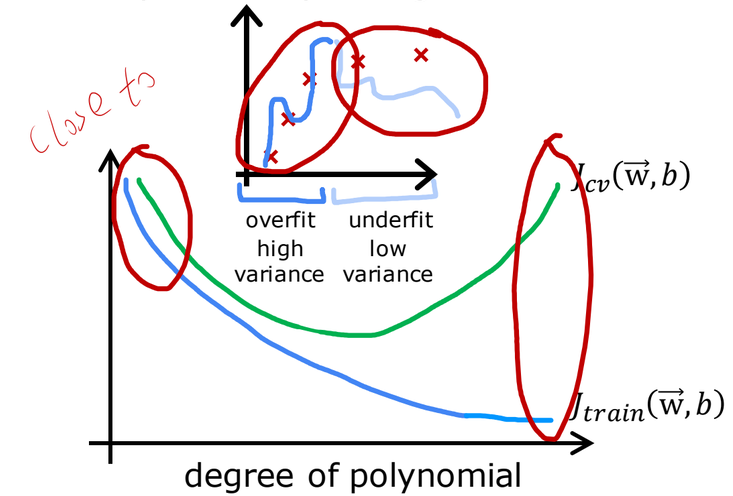
J_{cv} is high

Understanding bias and variance



Diagnosing bias and variance

How do you tell if your algorithm has a bias or variance problem?



High bias (underfit)

J_{train} will be high

($J_{train} \approx J_{cv}$)

High variance (overfit)

$J_{cv} \gg J_{train}$

(J_{train} may be be low)

High bias and high variance

J_{train} will be high

and $J_{cv} \gg J_{train}$



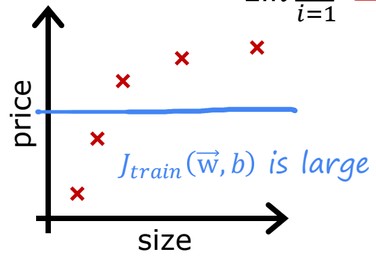
Bias and variance

Regularization and bias/variance

Linear regression with regularization

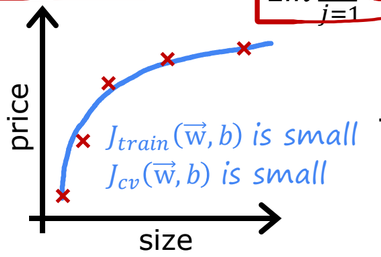
Model: $f_{\vec{w},b}(x) = w_1x + w_2x^2 + w_3x^3 + w_4x^4 + b$

$$J(\vec{w},b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)^2 + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2$$



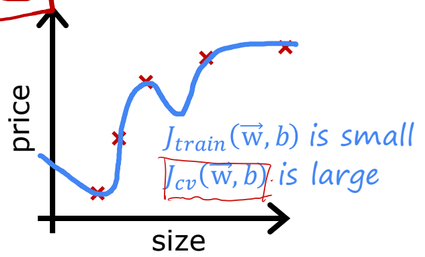
Large λ
High bias (underfit)

$\lambda = 10{,}000$ $w_1 \approx 0, w_2 \approx 0$
$f_{\vec{w},b}(\vec{x}) \approx b$



Intermediate λ

λ



Small λ
High variance (overfit)
$\lambda = 0$

Choosing the regularization parameter λ

Model: $f_{\vec{w},b}(x) = w_1x + w_2x^2 + w_3x^3 + w_4x^4 + b$

1. Try $\lambda = 0$ → $\min_{\vec{w},b} J(\vec{w},b)$ → $w^{<1>}, b^{<1>}$ → $J_{cv}(w^{<1>}, b^{<1>})$
2. Try $\lambda = 0.01$ → $\min_{\vec{w},b} J(\vec{w},b)$ → $w^{<2>}, b^{<2>}$ → $J_{cv}(w^{<2>}, b^{<2>})$
3. Try $\lambda = 0.02$ → $J_{cv}(w^{<3>}, b^{<3>})$
4. Try $\lambda = 0.04$
5. Try $\lambda = 0.08$ → $J_{cv}(w^{<5>}, b^{<5>})$

⋮

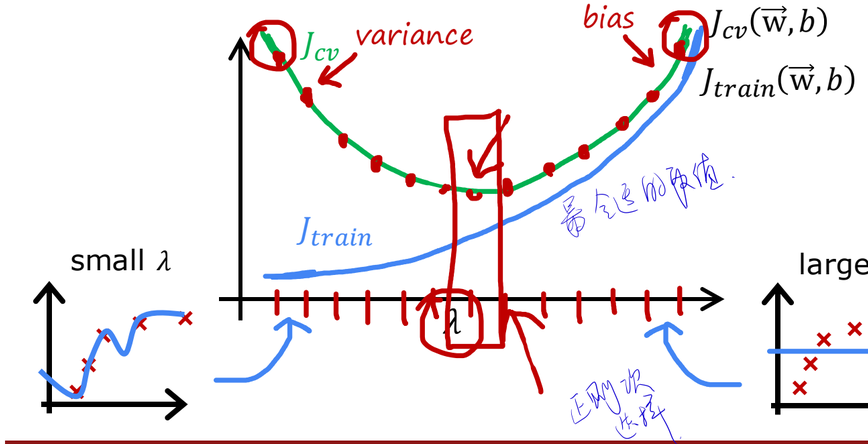
12. Try $\lambda \approx 10$ → $w^{<12>}, b^{<12>}$ → $J_{cv}(w^{<12>}, b^{<12>})$

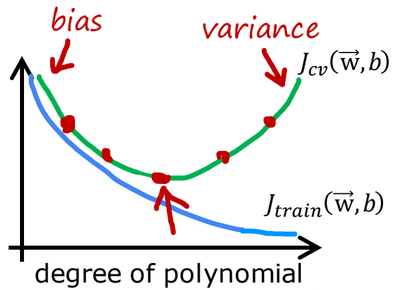
Pick $w^{<5>}, b^{<5>}$

Report test error: $J_{test}(w^{<5>}, b^{<5>})$

Bias and variance as a function of regularization parameter λ

$$J(\vec{w}, b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)^2 + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2$$







Bias and variance

Establishing a baseline level of performance

Speech recognition example



Human level performance : 10.6%

Training error J_{train} : 10.8%

Cross validation error J_{cv} : 14.8%

0.2%

4.0%

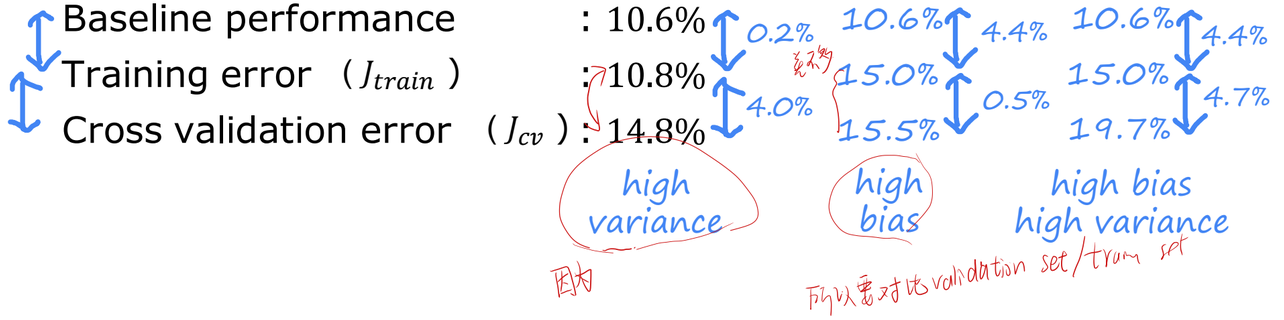


Establishing a baseline level of performance

What is the level of error you can reasonably hope to get to?

- Human level performance
- Competing algorithms performance (看竞争算法)
- Guess based on experience

Bias/variance examples





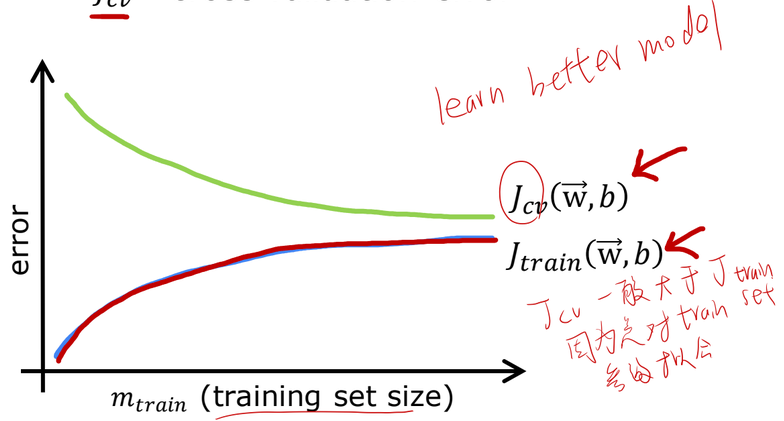
Bias and variance

Learning curves

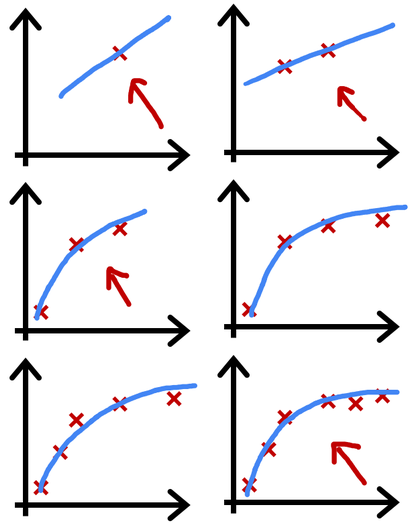
Learning curves

J_{train}= training error

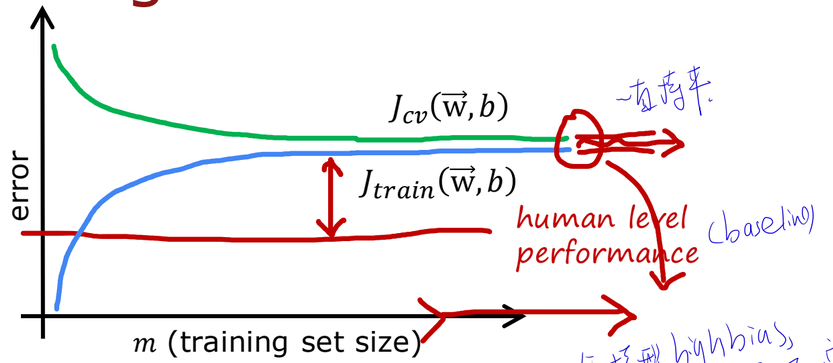
J_{cv}= cross validation error

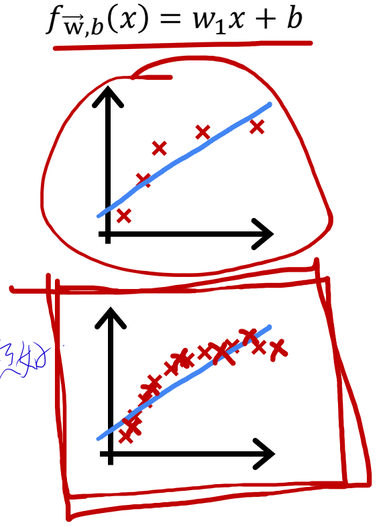


$f_{\vec{w},b}(x) = w_1 x + w_2 x^2 + b$



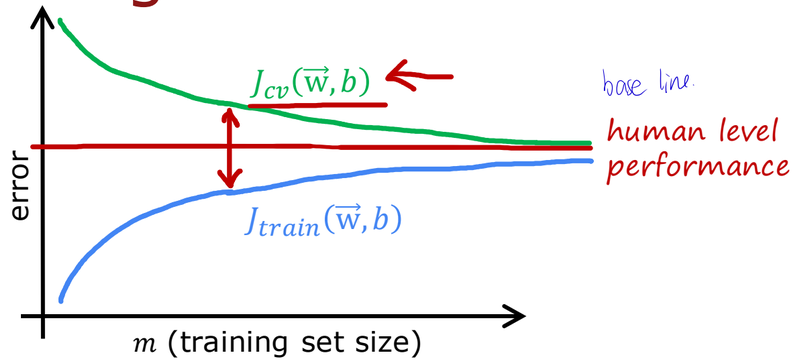
High bias





if a learning algorithm suffers from high bias, getting more training data will not (by itself) help much.

High variance → 偏差小 但没有很好泛化



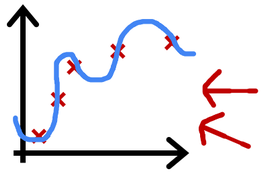
base line.

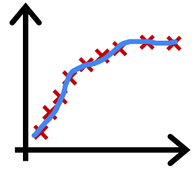
human level performance

if a learning algorithm suffers from high variance, getting more training data is likely to help.

$f_{\vec{w},b}(x) = w_1x + w_2x^2 + w_3x^3 + w_4x^4 + b$

(with small λ)







Bias and variance

Deciding what to try next revisited

Debugging a learning algorithm

You've implemented regularized linear regression on housing prices

$$J(\vec{w}, b) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)^2 + \frac{\lambda}{2m}\sum_{j=1}^{n} w_j^2$$

But it makes unacceptably large errors in predictions. What do you try next?

→ Get more training examples	fixes high variance
→ Try smaller sets of features $x, x^2, x^3, x^4, x^5 \ldots$	fixes high variance
→ Try getting additional features	fixes high bias
→ Try adding polynomial features $(x_1^2, x_2^2, x_1x_2, etc)$	fixes high bias
→ Try decreasing λ (步子跨小一点)	fixes high bias
→ Try increasing λ (步子跨大一点 (给更多灵活性))	fixes high variance



Bias and variance

Bias/variance and neural networks

The bias variance tradeoff

$f_{\vec{w},b}(x) = w_1x + b$

Simple model

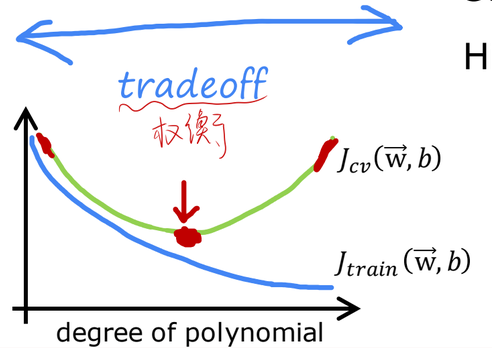
High bias

$f_{\vec{w},b}(x) = w_1x + w_2x^2 + b$

$f_{\vec{w},b}(x) = w_1x + w_2x^2 + w_3x^3 + w_4x^4 + b$

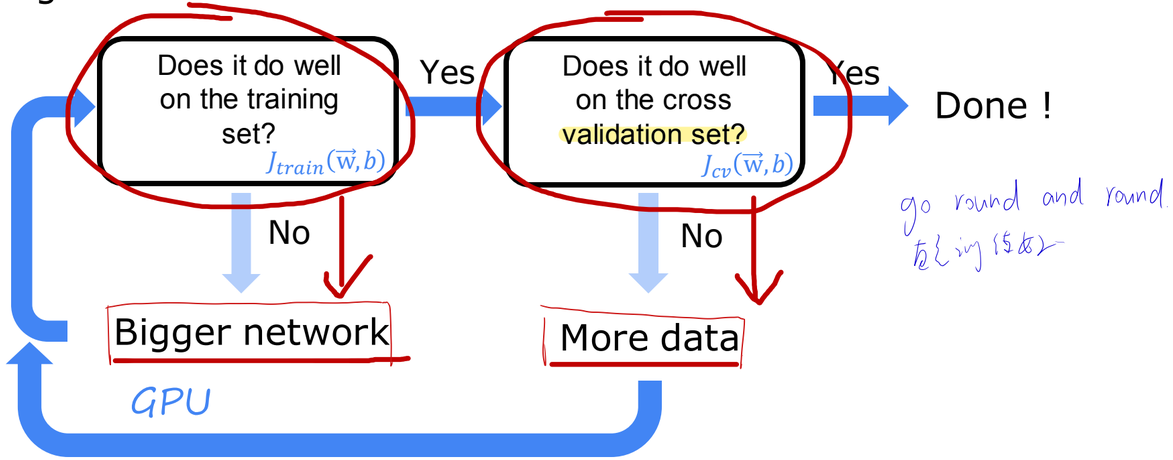
Complex model

High variance



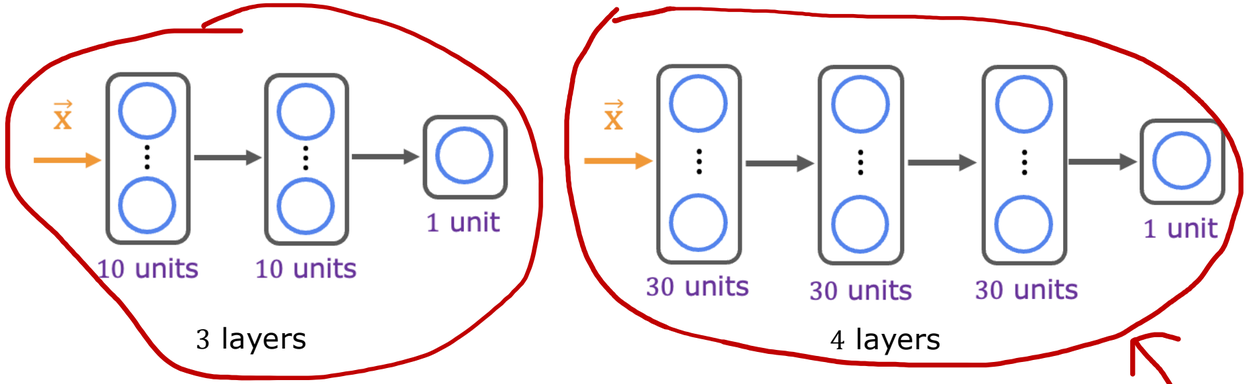
Neural networks and bias variance

Large neural networks are low bias machines



go round and round
抱怨ing 循环

Neural networks and regularization



A large neural network will usually do as well or better than a smaller one so long as regularization is chosen appropriately.

Neural network regularization

$$J(\mathbf{W},\mathbf{B}) = \frac{1}{m}\sum_{i=1}^{m} L\left(f\left(\vec{\mathrm{x}}^{(i)}\right), y^{(i)}\right) + \frac{\lambda}{2m}\sum_{\text{all weights } \mathbf{W}} \left(w^2\right)$$

b

Unregularized MNIST model

```
layer_1 = Dense(units=25, activation="relu")
layer_2 = Dense(units=15, activation="relu")
layer_3 = Dense(units=1, activation="sigmoid")
model = Sequential([layer_1, layer_2, layer_3])
```

Regularized MNIST model

```
layer_1 = Dense(units=25, activation="relu", kernel_regularizer=L2(0.01))
layer_2 = Dense(units=15, activation="relu", kernel_regularizer=L2(0.01))
layer_3 = Dense(units=1, activation="sigmoid", kernel_regularizer=L2(0.01))
model = Sequential([layer_1, layer_2, layer_3])
```

λ

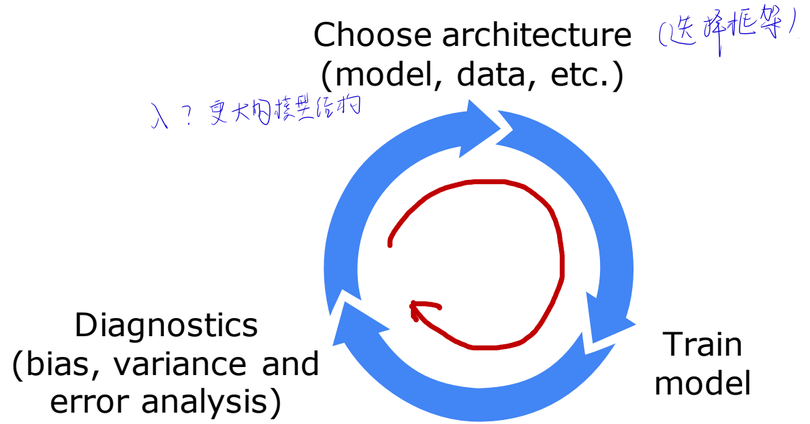
适当正则化，
能拥有一个更大的神经网络是可行的
但越深 → 算法变慢



Machine learning development process

Iterative loop of ML development

Iterative loop of ML development



Spam classification example

From: cheapsales@buystufffromme.com
To: Andrew Ng
Subject: Buy now!

Deal of the week! Buy now!
Rolex w4tchs - $100
Med1cine (any kind) - £50
Also low cost M0rgages
available.

可以开始考虑我们来
筛选器

From: Alfred Ng
To: Andrew Ng
Subject: Christmas dates?

Hey Andrew,
Was talking to Mom about plans
for Xmas. When do you get off
work. Meet Dec 22?
Alf

Building a spam classifier

Supervised learning: $\vec{x}$ = features of email
y = spam (1) or not spam (0)

Features: list the top 10,000 words to compute $x_1, x_2, \cdots, x_{10,000}$

$$\vec{x} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \\ 0 \\ \vdots \end{bmatrix} \begin{matrix} a \\ andrew \\ buy \\ deal \\ discount \\ \vdots \end{matrix}$$



```
From: cheapsales@buystufffromme.com
To: Andrew Ng
Subject: Buy now!

Deal of the week! Buy now!
Rolex w4tchs - $100
Med1cine (any kind) - £50
Also low cost M0rgages
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```

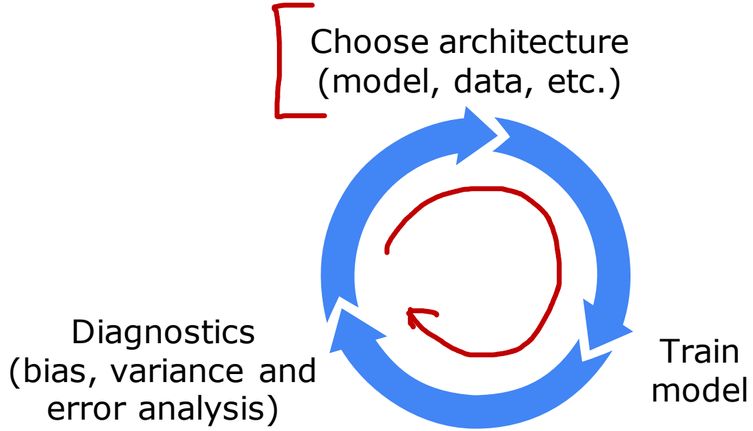
那我左边的问号是否出现，判断0或1。

Building a spam classifier

How to try to reduce your spam classifier's error?

- Collect more data. E.g., "Honeypot" project.
- Develop sophisticated features based on email routing (from email header).
- Define sophisticated features from email body. E.g., should "discounting" and "discount" be treated as the same word.
- Design algorithms to detect misspellings. E.g., w4tches, med1cine, m0rtgage.

Iterative loop of ML development





Machine learning development process

Error analysis

E sis

m_{cv} = ~~500~~ 5000 example set.

Algorithm misclass

Manually examine ategorize them
based on common

- Pharma: 21
- Deliberate mis
- Unusual email
- Steal passwor
- Spam messag

med1cine): 3

ge: 5

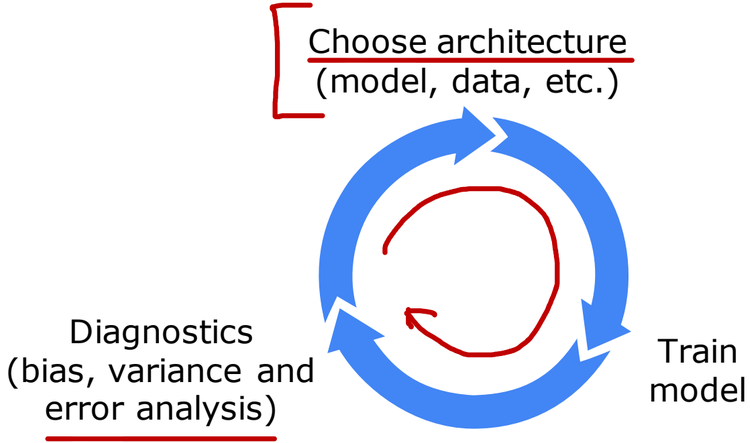
more data
features

Building a spam classifier

How to try to reduce your spam classifier's error?

- Collect more data. E.g., "Honeypot" project.
- Develop sophisticated features based on email routing (from email header).
- Define sophisticated features from email body. E.g., should "discounting" and "discount" be treated as the same word.
- Design algorithms to detect misspellings. E.g., w4tches, med1cine, m0rtgage.

Iterative loop of ML development





Machine learning development process

Adding data

Adding data

- Add more data of everything. E.g., “Honeypot” project.
- Add more data of the types where error analysis has indicated it might help.

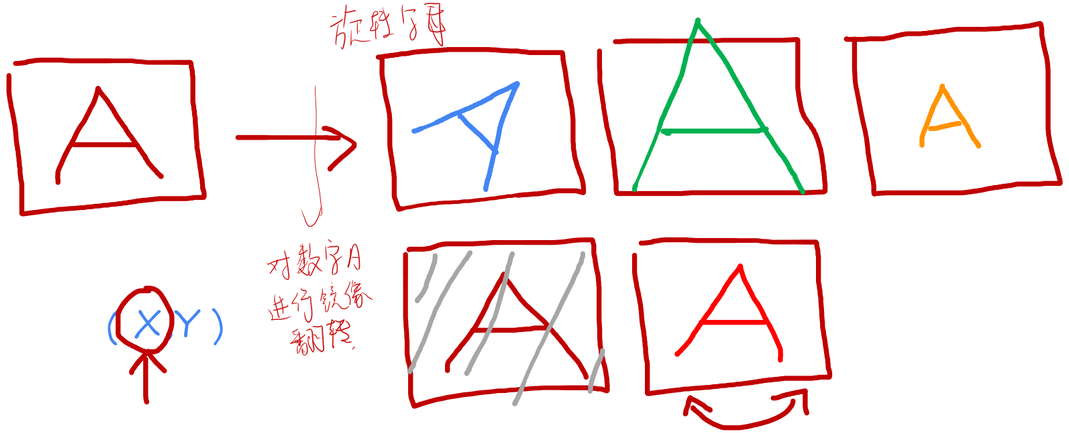
  Pharma spam

  E.g., Go to unlabeled data and find more examples of Pharma related spam.

Data augmentation

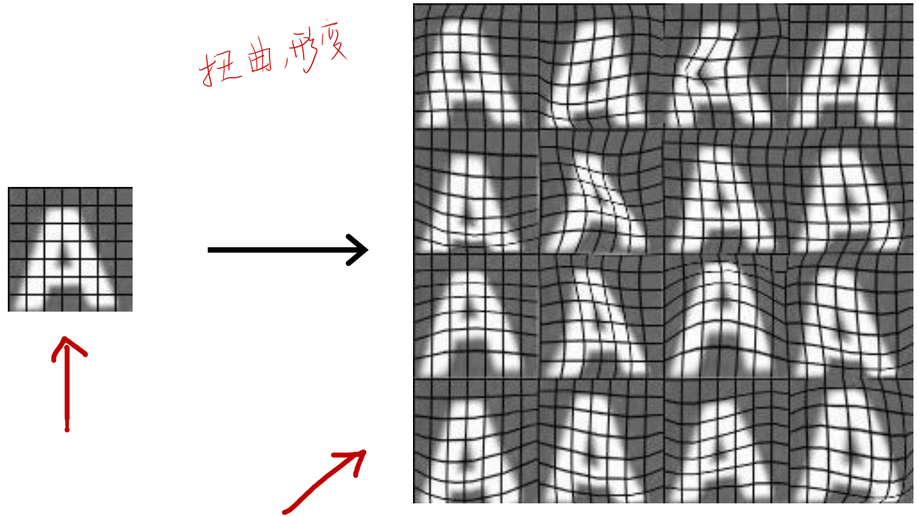
数据增强

Augmentation: modifying an existing training example to create a new training example.



Data augmentation by introducing distortions

扭曲，形变



Data augmentation for speech

Speech recognition example



Original audio (voice search: "What is today's weather?")



+ Noisy background: Crowd

增加了嘈杂的背景音



+ Noisy background: Car

变音



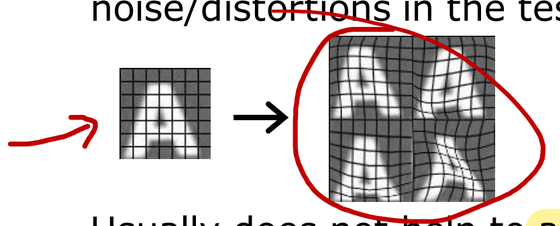
+ Audio on bad cellphone connection

信号不好

增强(数据加噪声

Data augmentation by introducing distortions

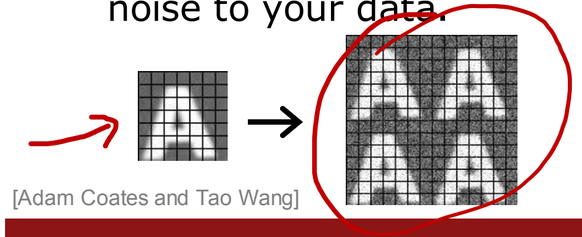
Distortion introduced should be representation of the type of noise/distortions in the test set.



Audio:
Background noise,
bad cellphone connection

Usually does not help to add purely random/meaningless noise to your data.

尽量用有意义的噪声



[Adam Coates and Tao Wang]

x_i=intensity (brightness) of pixel i

$x_i \leftarrow x_i$ + random noise

Data synthesis

Synthesis: using artificial data inputs to create a new training example.

Artificial data synthesis for photo OCR



[http://www.publicdomainpictures.net/view-image.php?image=5745&picture=times-square]

Artificial data synthesis for photo OCR



Real data



[Adam Coates and Tao Wang]

Artificial data synthesis for photo OCR



Real data

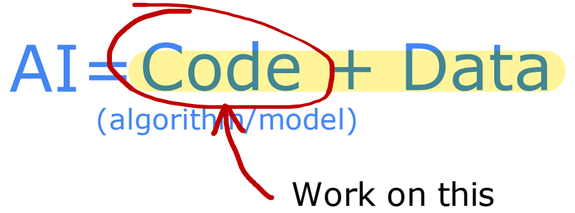


Synthetic data

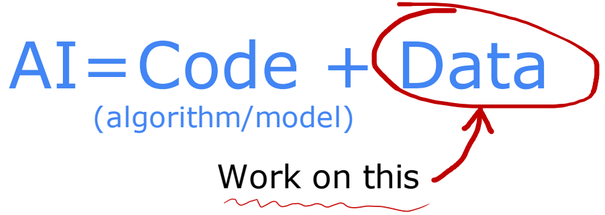
[Adam Coates and Tao Wang]

Engineering the data used by your system

Conventional model-centric approach:

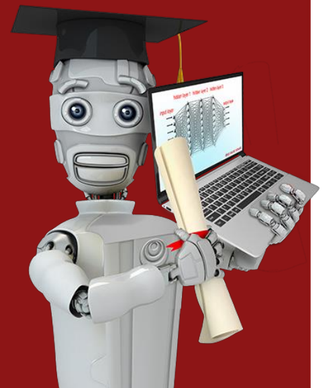


Data-centric approach:



DeepLearning.AI

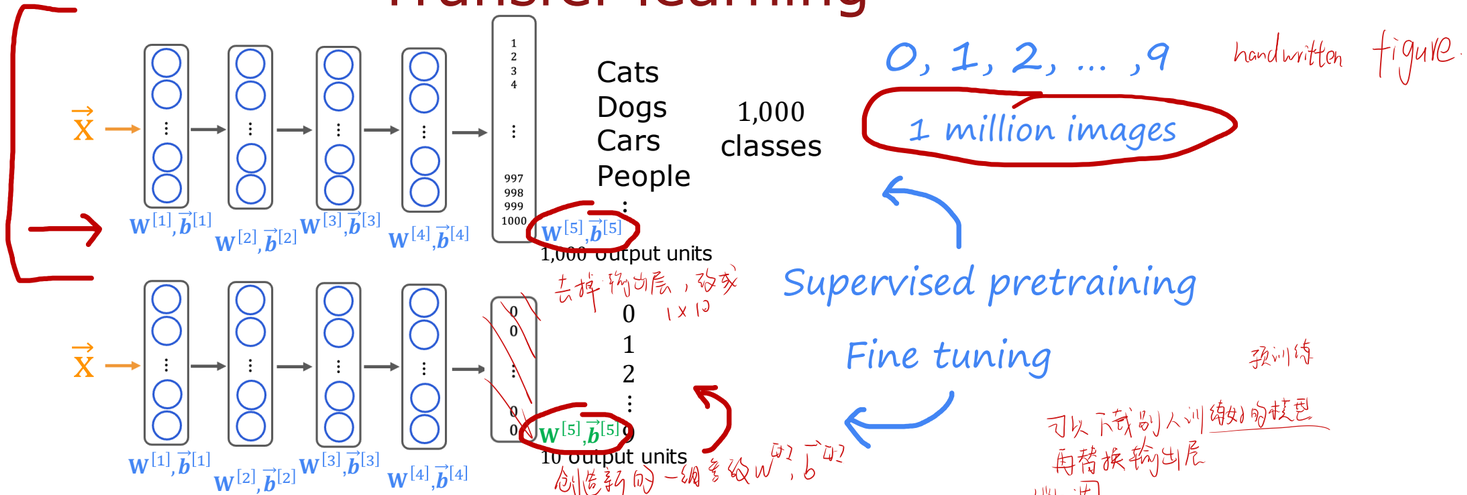
Stanford
ONLINE



Machine learning development process

Transfer learning: using data from a different task

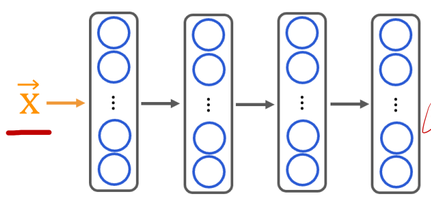
Transfer learning



Option 1: only train output layers parameters.

Option 2: train all parameters.

Why does transfer learning work?



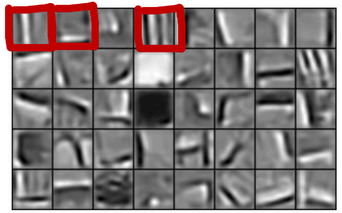
intuition behind 计
不同层检测不同特征

use the same input type

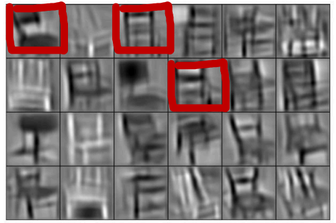
detects edges

detects corners

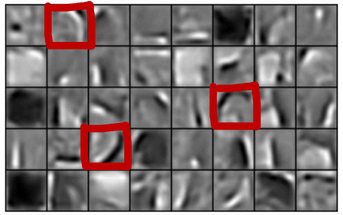
detects curves/basic shapes



Edges



Corners

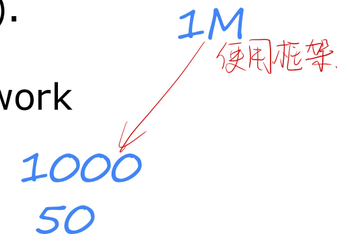


Curves / basic shapes

Transfer learning summary

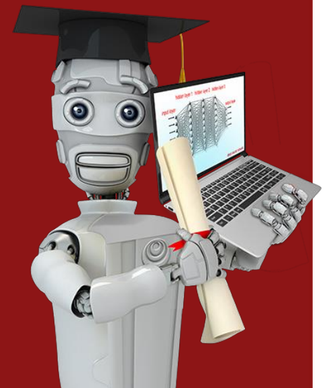
1. Download neural network parameters pretrained on a large dataset with same input type (e.g., images, audio, text) as your application (or train your own).

2. Further train (fine tune) the network on your own data.



DeepLearning.AI

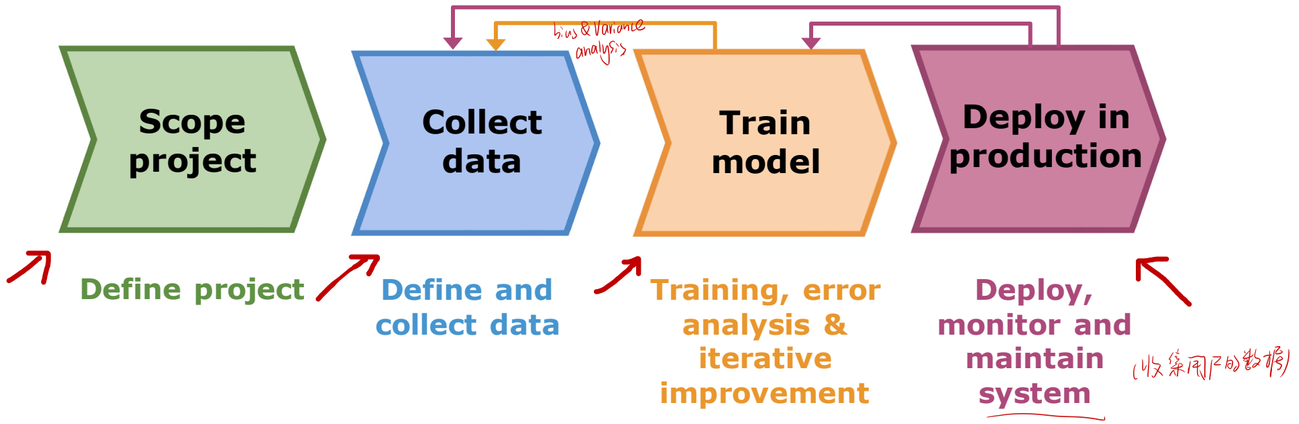
Stanford
ONLINE



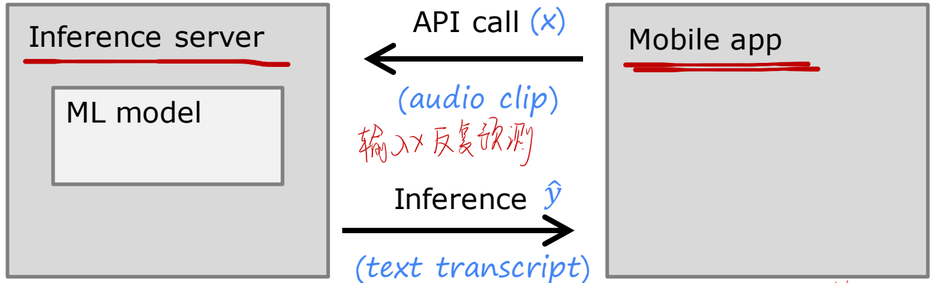
Machine learning development process

Full cycle of a machine learning project

Full cycle of a machine learning project



Deployment



→ Software engineering may be needed for:

- Ensure reliable and efficient predictions
- Scaling
- Logging
- System monitoring
- Model updates

部署，software engineering

MLOps

machine learning operations

机器学习操作



Machine learning development process

Fairness, bias, and ethics

Bias

Hiring tool that discriminates against women.

Facial recognition system matching dark skinned individuals to criminal mugshots.

Biased bank loan approvals.

Toxic effect of reinforcing negative stereotypes.

Adverse use cases

Deepfakes

传播有毒观点

Spreading toxic/incendiary speech through optimizing for engagement.

发布虚假评论

Generating fake content for commercial or political purposes.

Using ML to build harmful products, commit fraud etc.

Spam vs anti-spam : fraud vs anti-fraud.

Guidelines

Get a diverse team to brainstorm things that might go wrong, with emphasis on possible harm to vulnerable groups.

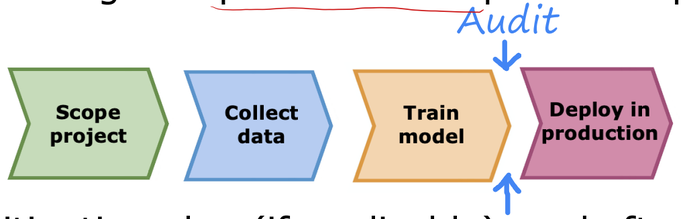
多元化团队，考虑弱势群体

Carry out literature search on standards/guidelines for your industry.

文献研究

Audit systems against possible harm prior to deployment.

潜在伤害



Develop mitigation plan (if applicable), and after deployment, monitor for possible harm.



Skewed datasets (optional)

Error metrics for skewed datasets

Rare disease classification example

Train classifier $f_{\vec{w},b}(\vec{x})$ ($y = 1$ if disease present, $y = 0$ otherwise)

Find that you've got 1% error on test set
(99% correct diagnoses) 二元分类

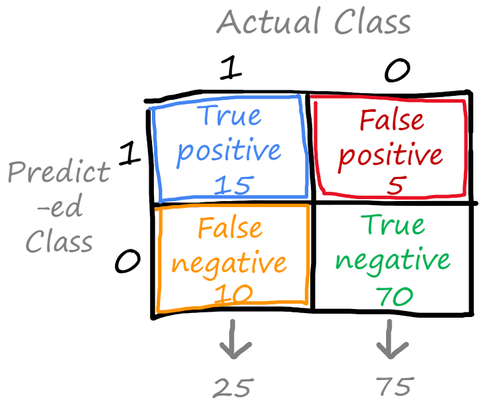
Only 0.5% of patients have the disease

```
print("y=0")
```

99.5% accuracy, 0.5% error ←
1%
1.2%

Precision/recall

$y = 1$ in presence of rare class we want to detect.



Precision:
(of all patients where we predicted $y = 1$, what fraction actually have the rare disease?)

$$\frac{\text{True positives}}{\#\text{predicted positive}} = \frac{\text{True positives}}{\text{True pos} + \text{False pos}} = \frac{15}{15+5} = 0.75$$

Recall:
(of all patients that actually have the rare disease, what fraction did we correctly detect as having it?)

$$\frac{\text{True positives}}{\#\text{actual positive}} = \frac{\text{True positives}}{\text{True pos} + \text{False neg}} = \frac{15}{15+10} = 0.6$$

```
print("y=0")
```



Skewed datasets (optional)

Trading off precision and recall

high precision — 如果诊断出
某种疾病，你有可能患有这种疾病

Trading off precision and recall

Logistic regression: $0 < f_{\vec{w},b}(\vec{x}) < 1$

Predict 1 if $f_{\vec{w},b}(\vec{x}) \geq 0.5$ ~~0.7~~ ~~0.9~~ 0.3

Predict 0 if $f_{\vec{w},b}(\vec{x}) < 0.5$ ~~0.7~~ ~~0.9~~ 0.3

$$\text{precision} = \frac{\text{true positives}}{\text{total predicted positive}}$$

$$\text{recall} = \frac{\text{true positives}}{\text{total actual positive}}$$

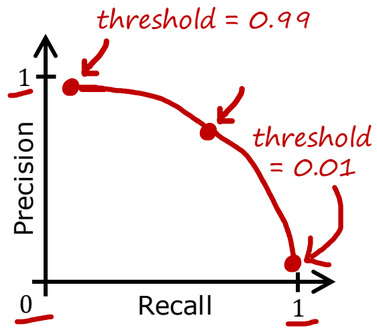
Suppose we want to predict $y = 1$ (rare disease) only if very confident.

→ higher precision, lower recall.

Suppose we want to avoid missing too many cases of rare disease (when in doubt predict $y = 1$)

→ lower precision, higher recall.

More generally predict 1 if: $f_{\vec{w},b}(\vec{x}) \geq$ threshold.



F1 score

How to compare precision/recall numbers?

	Precision (P)	Recall (R)	Average	F_1 score
Algorithm 1	0.5	0.4	0.45	0.444
Algorithm 2	0.7	0.1	0.4	0.175
Algorithm 3	0.02	1.0	0.501	0.0392

不能直接取平均值

只能从两种算法中选择一个进行下一步操作

```
print("y=1")
```

$$\text{Average} = \frac{P+R}{2}$$

$$\text{F1 score} = \frac{1}{\frac{1}{2}\left(\frac{1}{P}+\frac{1}{R}\right)} = 2\frac{PR}{P+R}$$

计算方式取个F1-Score高的